Sustainability Lightning Talks



OCP FUTURE TECHNOLOGIES SYMPOSIUM

October 14, 2026
San Jose, California

A Digital-Twin-Based Framework for Data Center Sustainability Retrofit Evaluation

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Hitachi America, Ltd.



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Background & Motivation

Increasing rack power density to support AI workloads.

Legacy air-cooling in existing data center infrastructure becomes increasingly constrained.

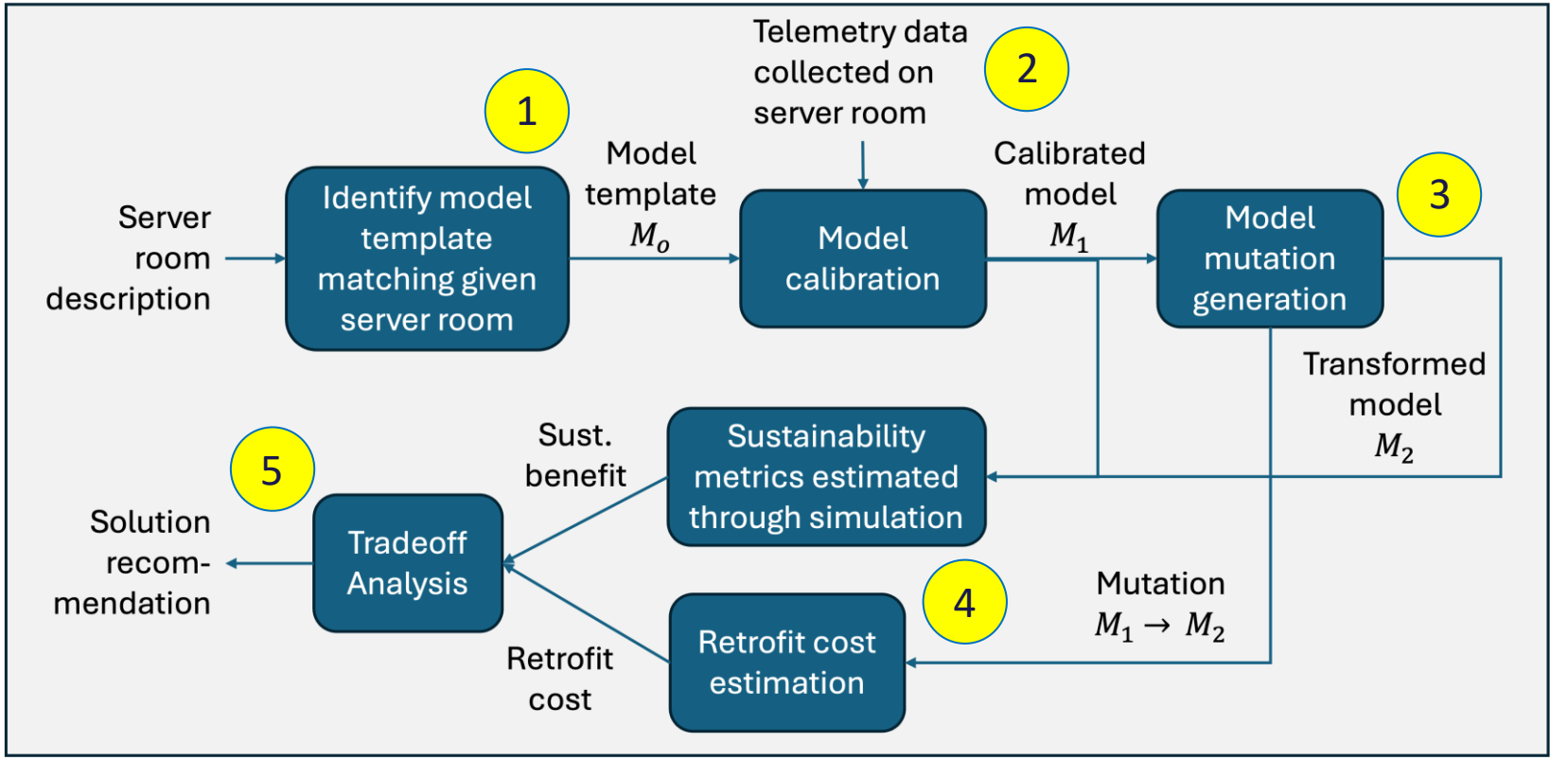
With many possible retrofit choices, challenge for data center operators: which retrofit provides the best sustainability-cost tradeoff?

Operators need a systematic way to compare retrofit alternatives before investing.

Simulation enables objective retrofit evaluation

Digital-Twin-Based Framework Overview

Automatically generate and evaluate retrofit designs



End-to-End Workflow

Telemetry-Driven Digital Twin Calibration

Calibration makes the digital twin facility-specific

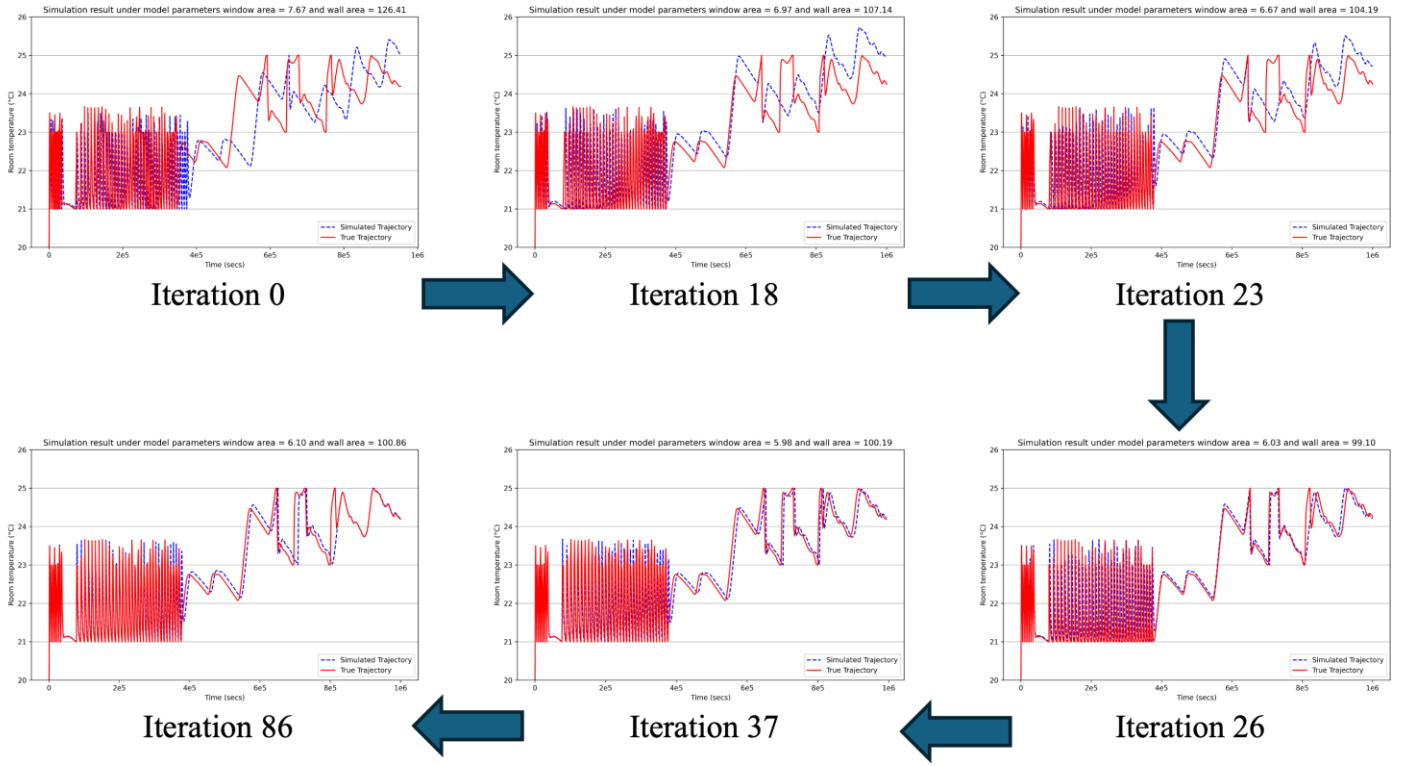
Template
Reusable parameterized model

Telemetry
Historical facility measurements

Calibration
Adjust unknown parameters to match observations

Result
Facility-specific digital twin

Large digital twins
Scalable sequential optimization algorithm



Matching Simulated and observed trajectories

Mutation-based Digital Twin Retrofit Generation

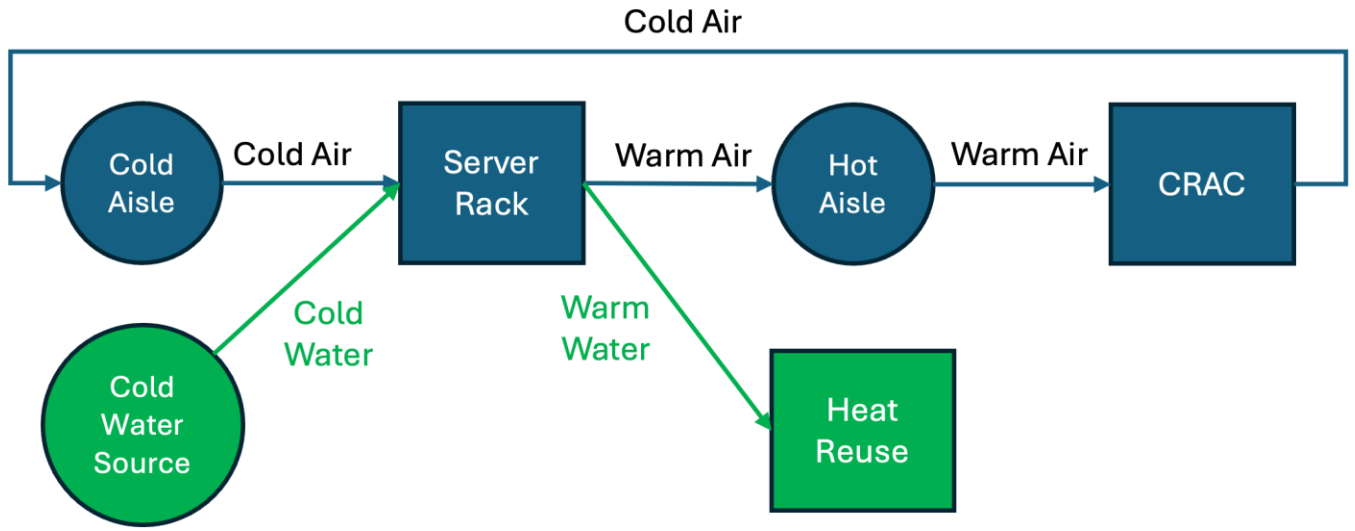
Graph mutations explore the retrofit design space

Digital twin as a graph

- **Nodes**: facility components
- **Edges**: connections and flows

Graph mutations

- **Insert**
- **Replace**
- **Modify**
- **Compound transformation**



Example compound mutation that adds liquid cooling to an air-cooled rack

Illustrative Data Center Facility Retrofit Scenario

Retrofit #1 provides the best sustainability-cost tradeoff

Baseline configuration

Air-cooled server room, 100 kW AI rack, CRAC

Retrofit Option #1

Adds direct liquid cooling, warm coolant leaving at 60°C feeds an organic Rankine cycle (ORC) generator

Retrofit Option #2

Additionally incorporates a heat pump to raise warm coolant output to 80°C feeding an ORC

Framework reduces a large retrofit design space to a quantitative sustainability/economic comparison

	Baseline	Retrofit #1	Retrofit #2
PUE	1.90	**1.20**	1.21
Retrofit Cost	--	**$350k**	$500k
Payback	--	**4.5 yrs**	6.3 yrs
WHRF	0.00	0.04	**0.06**

Conclusion

Telemetry-calibrated digital twins

Model existing data center facilities

Automated retrofit exploration

Generate and simulate many candidate configurations

Quantitative decision support

Compare sustainability benefits against implementation costs

A digital twin enables data-driven retrofit decisions

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Water-Based Two-Phase Cooling

Boya Qu

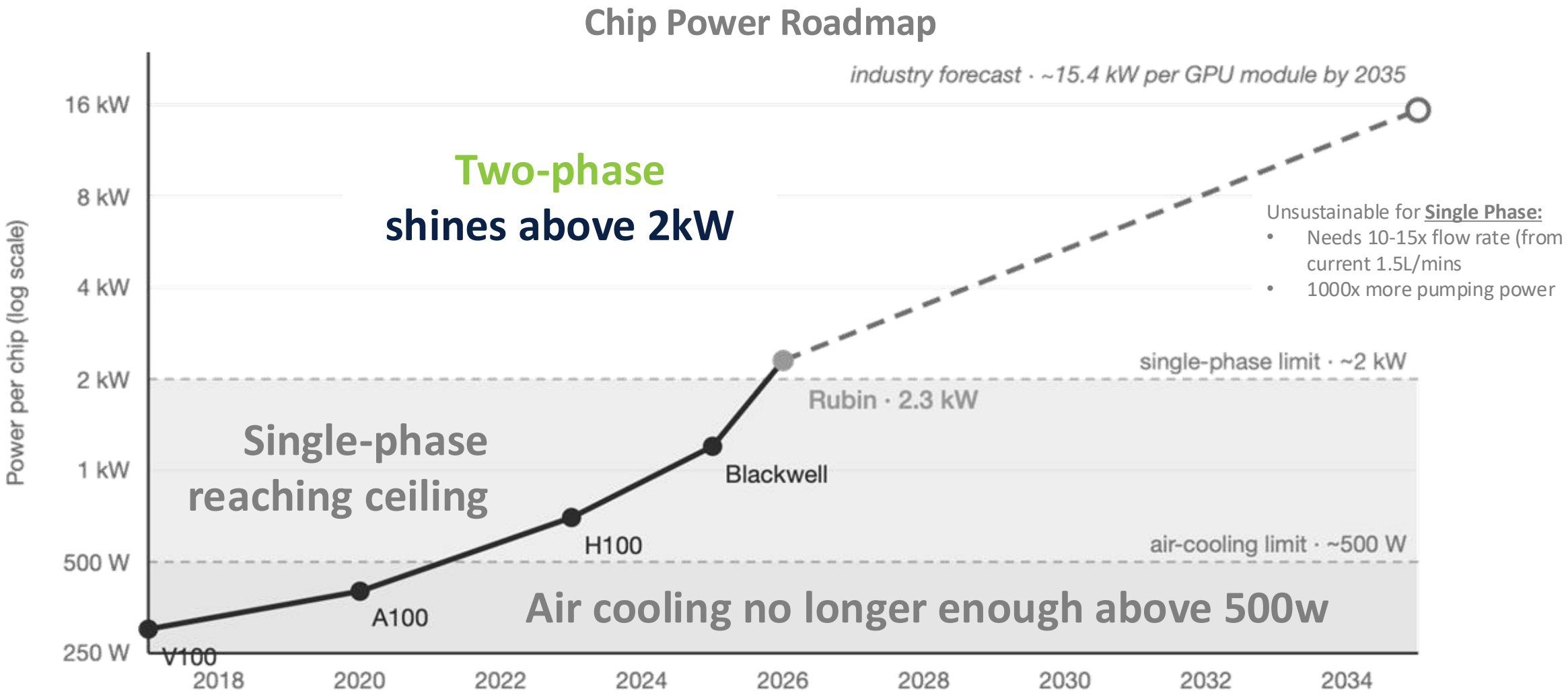
CEO, SemiValley Technologies, Inc.



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Current Liquid Cooling Maxing Out



Introducing | Two-Phase, Water-Based Cold Plate

1 Jet Impingement

Continuously re-wets hotspots via jet arraysand clears vapor from the die

2 Two Phase with Separation

Leverage latent heat; Separates liquid and vapor paths to suppress blockage, oscillation, and dry-out.

3 Using Water vs. PFAS

Uses sub-atmospheric water boiling at ~60°C: no PFAS.

MEASURED RESULTS SemiValley

Performance

- 328 W/cm² and 0.1 L/min
- 125 kW/m²·K HTC
- 0.0098 K/W liquid-side Rth
- 50.8% less temperature oscillation (delta from 4.76°C → 0.44 °C)
- 23% lower ΔP (Pressure oscillation)
- 300 W NVIDIA V100 @ 0.15 L/min

Stability for deployment

High Heat Flux, Low Flow, and Stable

BACKGROUND

NVIDIA Inception Program

ARPA-E

UCLA

intel connection

OPEN Compute Project

Datacenter Scale | Economics in Deployment

1	**LOWER ENERGY COSTS**	**$34 M USD Power Savings**	Electricity saved per year at 100MW of IT load, against an air-cooled baseline, $7.5M against best-in-class liquid-cooled 1.15 PUE
2	**MORE COMPUTE, MORE TOKENS**	**+42 % More Tokens**	For Anthropic inference tokens, that means $500M additional revenue for a 100MW site
3	**SIGNIFICANTLY LESS WATER**	**4x Water Reduction**	Saving water with 1/ 4 flow rate reduction, for about - 1.6 bn L / year
4	**GREATER THERMAL STABILITY**	**10x Less Temp. Oscillation**	Surface-temp oscillation fell from 4.76 to 0.44 °C, measured on identical hardware

For 100 MW of IT load

Path to Deployment | Validation & Collaboration

CURRENT PRIORITIES

Single-chip Vera Rubin Ultra + Feynman level
Testing 2000w and 5000w TTV performance

Multi-chip Performance
Testing multi-chip at high power

CDU Validation PoCs
Working with several CDU companies on existing commercial CDU integration

STILL OPEN

Rack-level behavior
Flow distribution, vapor handling and CDU response across a full rack with manifold design

Facility-scale PUE
1.05 is an engineering target , will measure in real deployment

Long-Term Reliability
Ongoing evaluation of corrosion, erosion, and surface stability under extended operating conditions

INDSUTRY COLLABORATION

CDU /Cooling Integrators
With major CDU/ industry integrators companies

Server and Power OEMs
Multi-chip tray integration with manifold design & control systems

Datacenter Operators
Partner site for rack-level demonstration, and a facility-scale PUE measurements

Joint Reference Design & Co-Author standards
OCP and ASHRAE spec work for water two-phase, plus independent long-term materials data

PATH TO DEPLOYMENT

Q3 2026	Q4 2026	Q1 2027	Q2 2027	Q3 2027
Single Chip	**Single Tray** **Multi – Tray Rack**	**Full Rack Validation**	**Multi - Rack Validation** **Cluster / Modular Testing**	**Product Pre-Sale**

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Towards Sustainable AI Data Centers Through Direct Waste Heat Recovery Enabled by Membrane-Assisted Heat Sink Technology

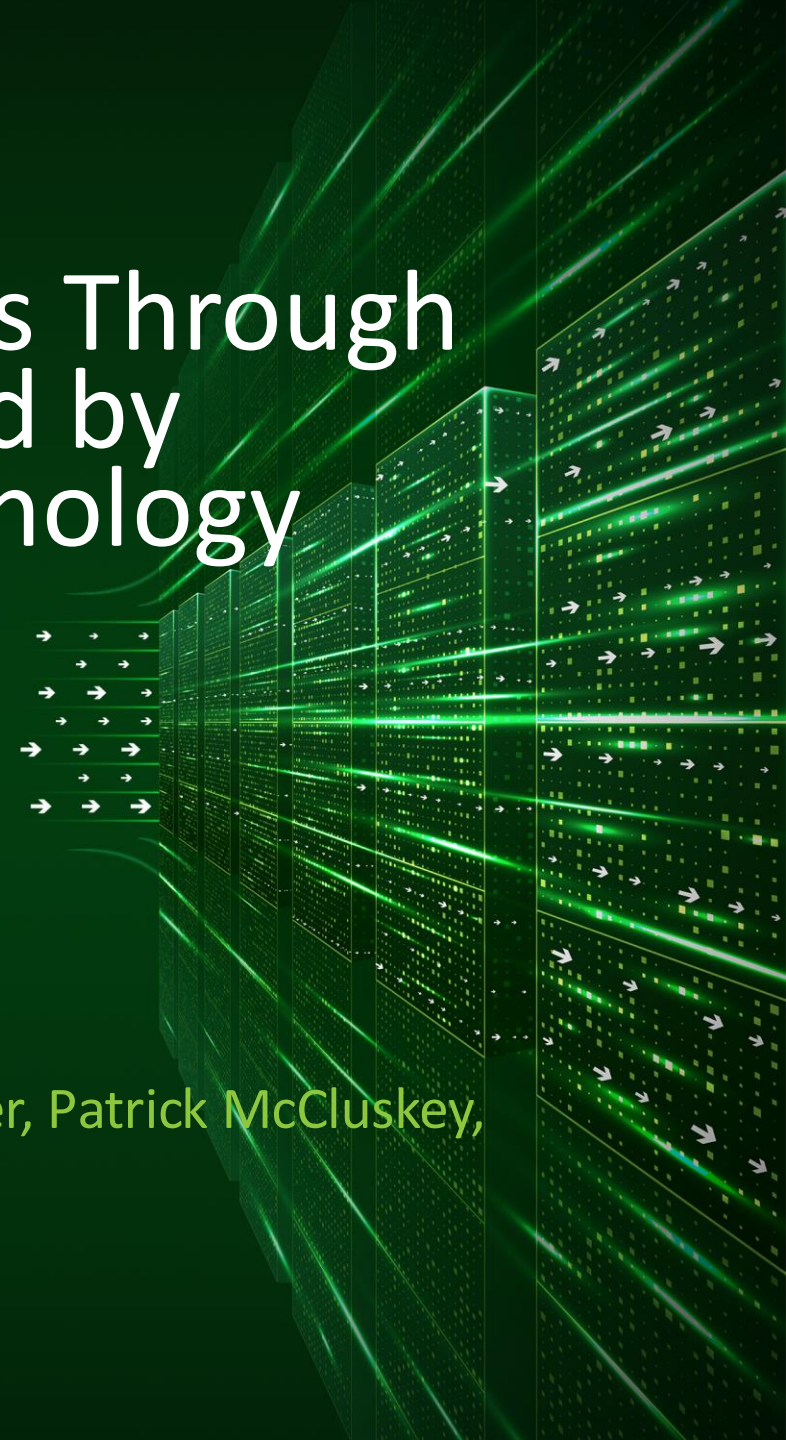
Saeed Moghaddam

Professor, University of Florida

Roshith Mittakolu

Graduate Research Assistant, University of Maryland, College Park

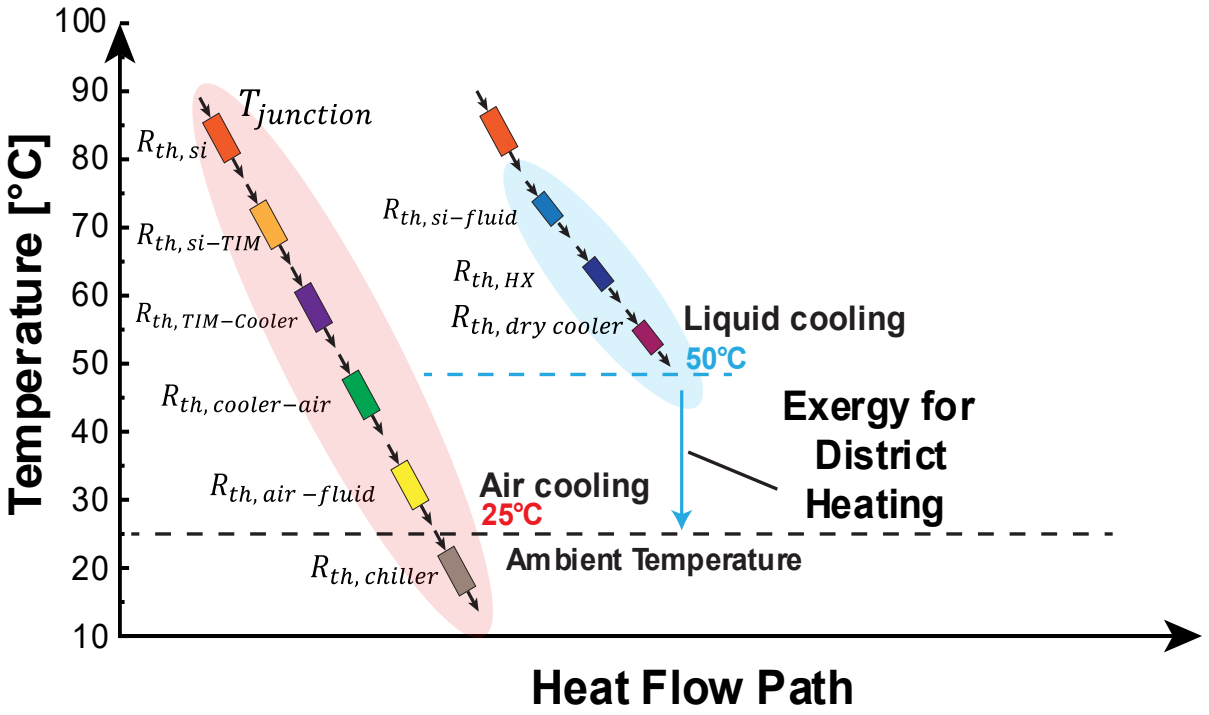
Team: Roshith Mittakolu, Ibaad Gandikota, Damena Agonafer, Patrick McCluskey, Saeed Moghaddam

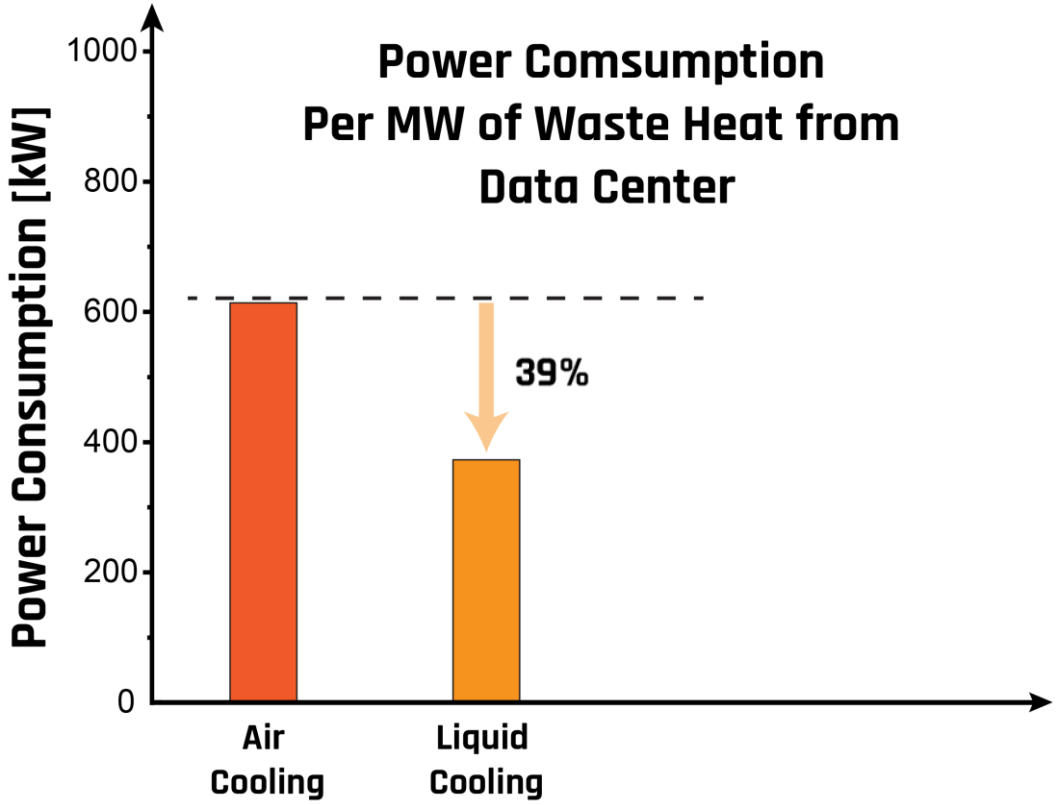


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Core Challenge with Waste Heat Recovery





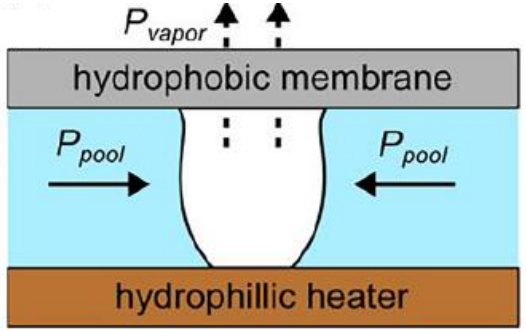
Current State of the art Data Centers

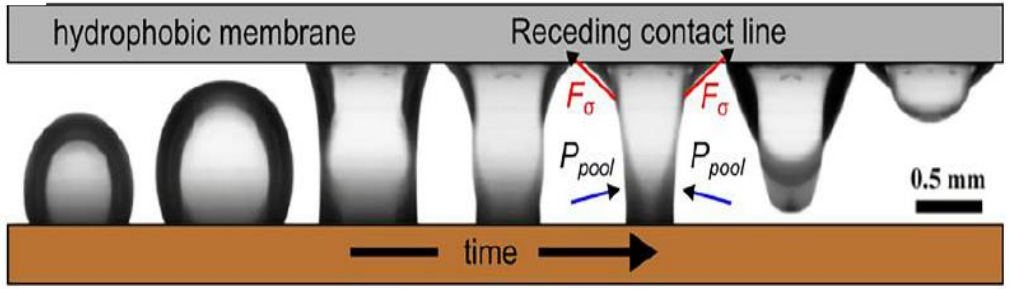
Use air or chilled-water cooling architectures and reject heat at low temperatures (~30-50 °C).

Low outlet temperatures translate to low-grade heat

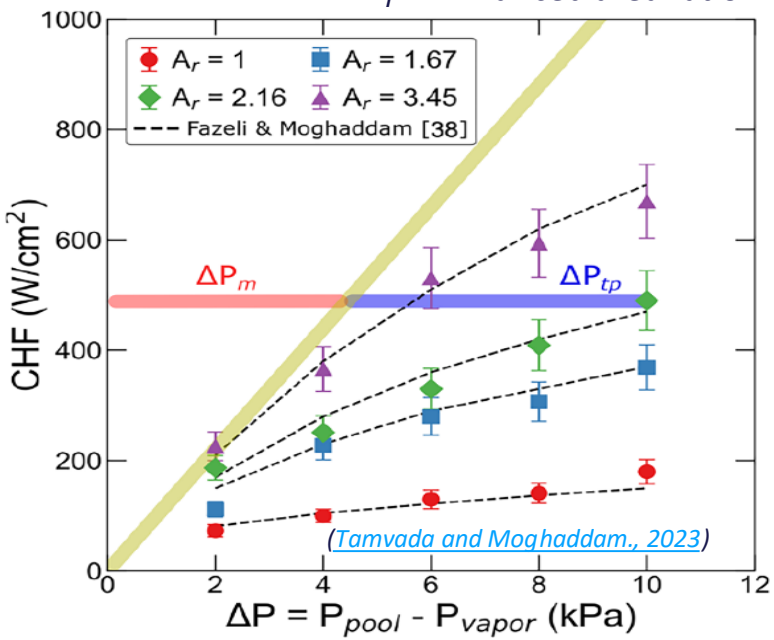
This makes waste heat reuse inefficient or economically unattractive.

Our MHS Technology – No PFAS Refrigerants, No Maldistribution, Achieves exit quality of 1





A_r = Enhanced area ratio



(Tamvada and Moghaddam., 2023)

Membrane heat sink (MHS) achieves exit vapor quality one, while dissipated ultra high heat fluxes and uses water as the working fluid resulting in very efficient operation of the system

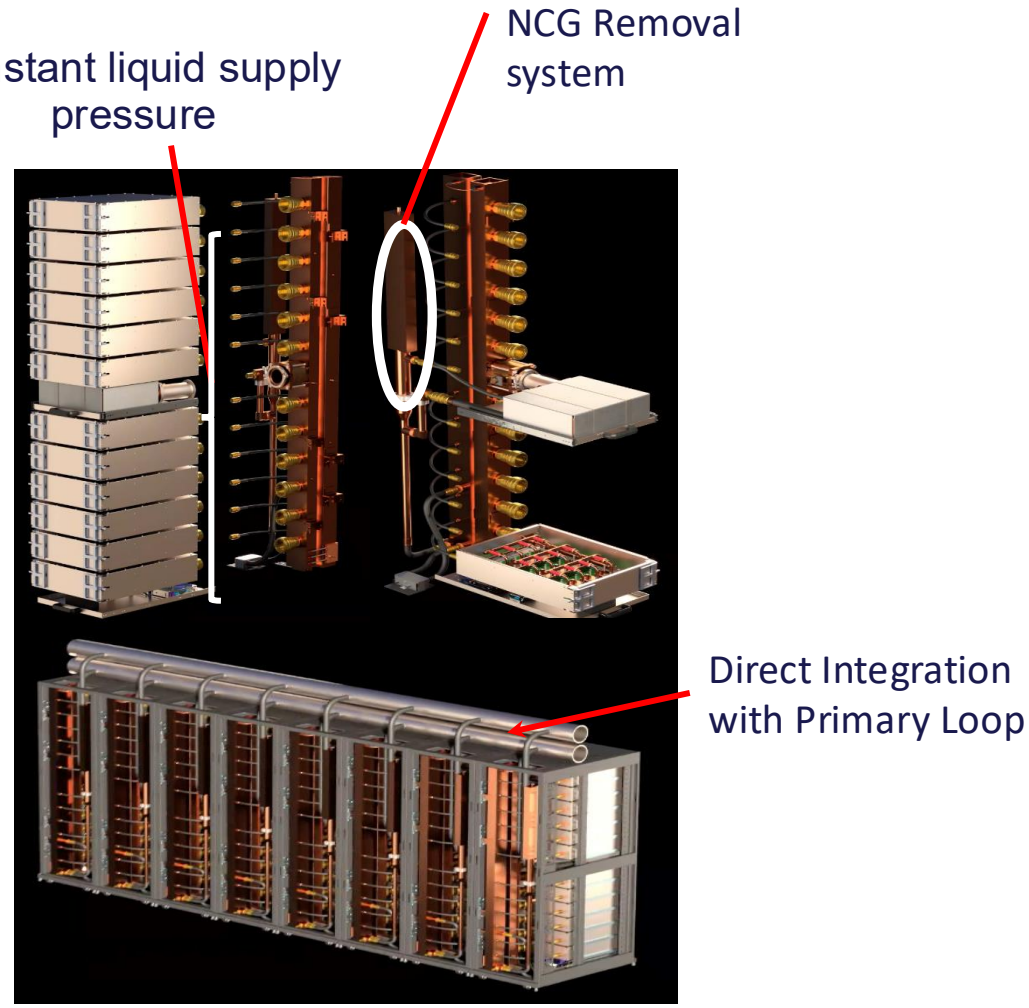
Property	Units	TMC-7300E	Opteon 2P50	Opteon SF33	Water
Dielectric Strength	kV/mm	>25	41	10	-
Specific Heat (c_p)	kJ/(kgK)	1140	1090	1200	4180
Thermal Conductivity (k_f)	W/(mK)	0.063	0.073	0.077	0.608
Latent Heat (L)	J/g	102	108	166	2260

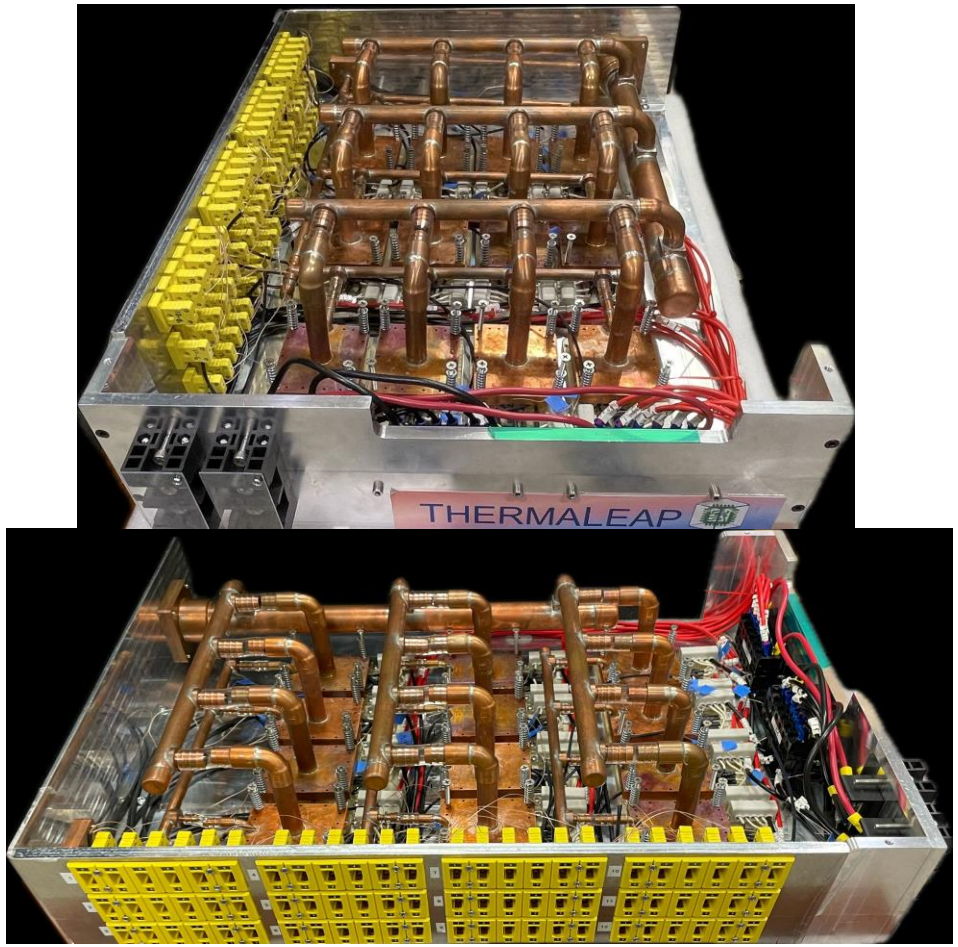
→ ~4x higher specific heat

→ ~10x higher thermal conductivity

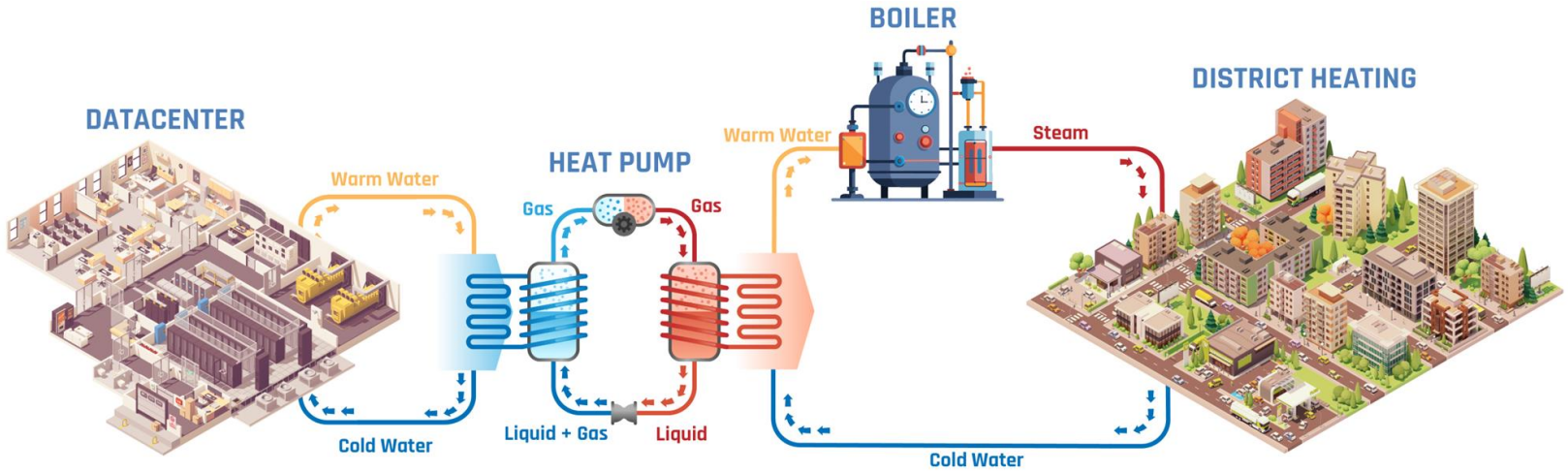
→ ~20x higher latent heat

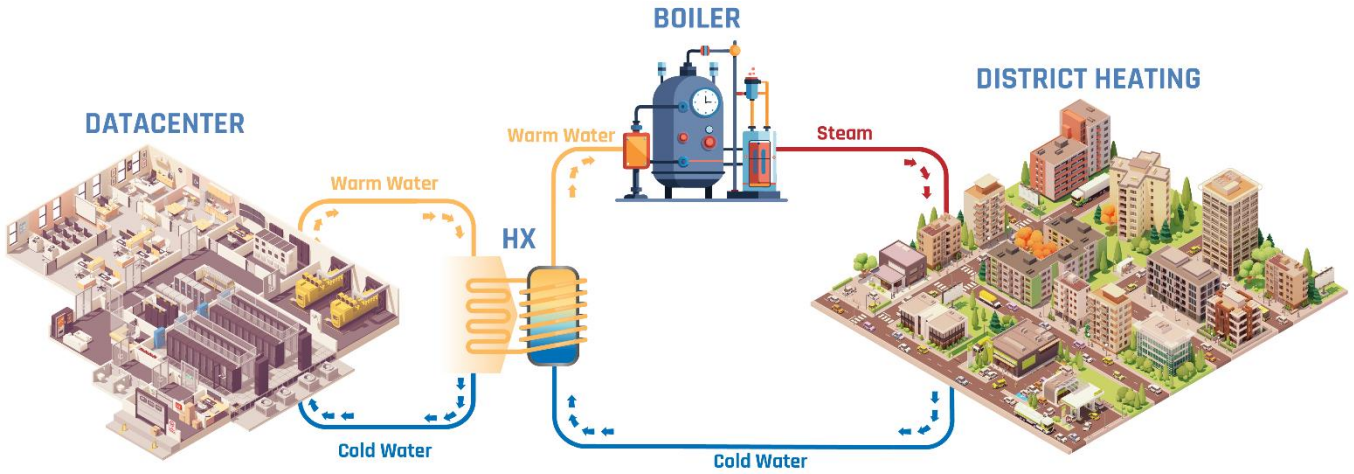
Rack Architecture and Implementation in Data Centers

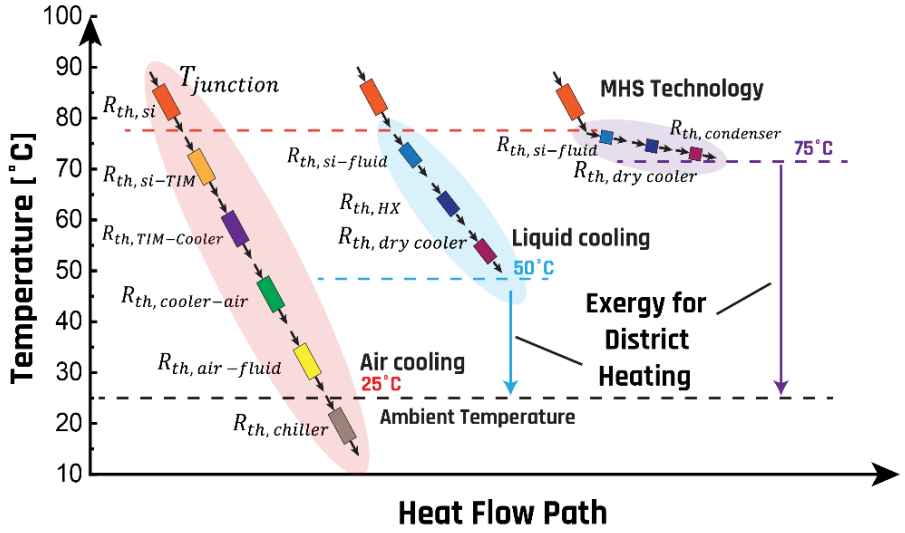


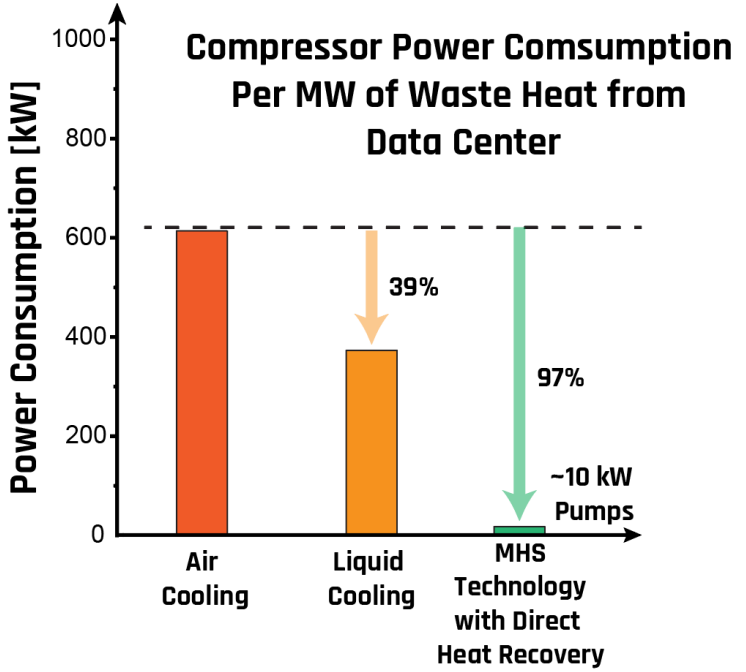


Direct Waste Heat Recovery









Transforming AI Data Centers: From Social Burden to Social Benefit

Next Steps

Scalable Deployment

- ✓ Mass manufacturable cold plates
- ✓ Standardized rack integration

Strategic Collaboration

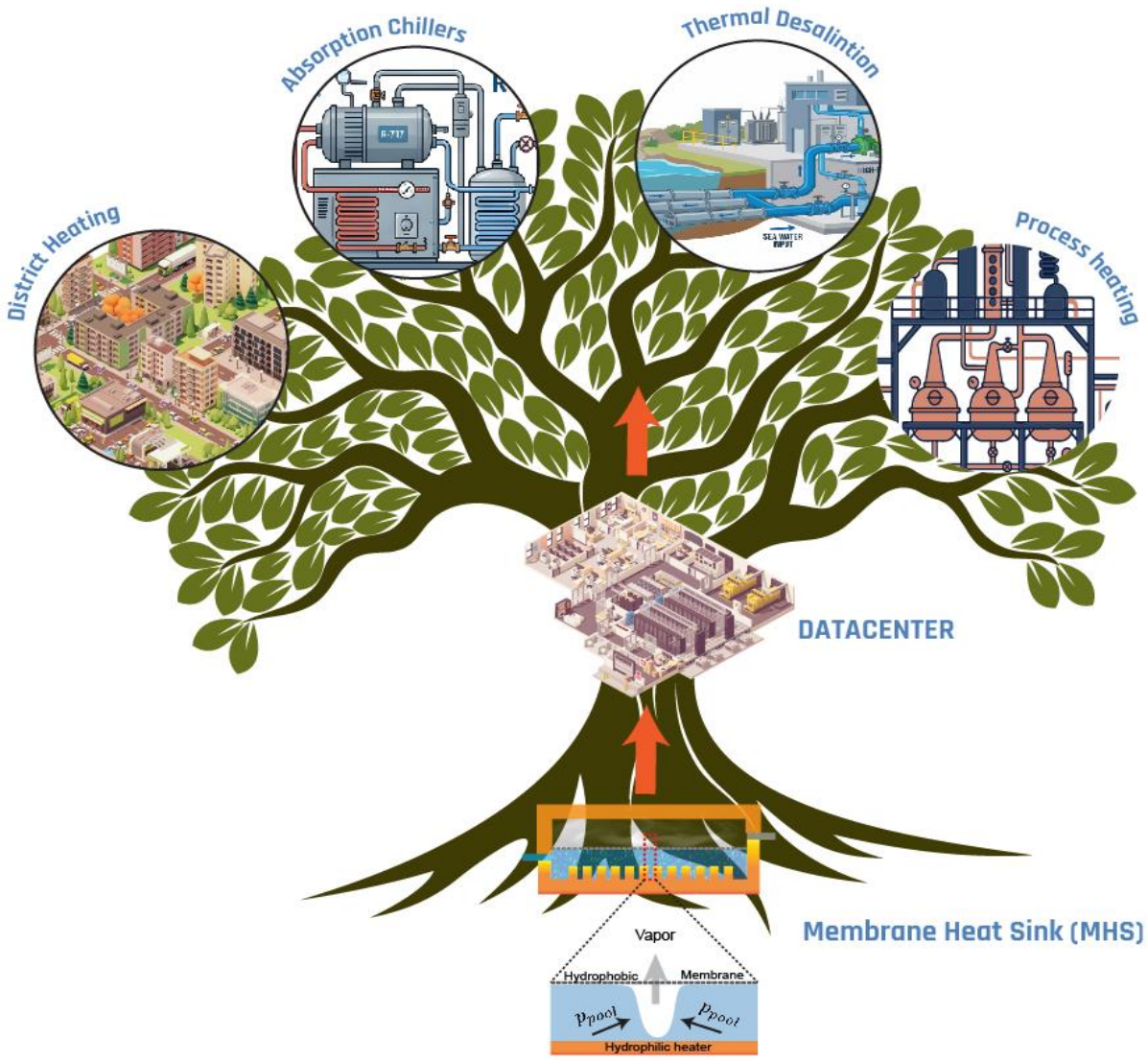
- ✓ Collaboration with potential colocation providers or hyper-scalers.

Infrastructure Integration

- ✓ Integration of MHS retrofitted racks with waste heat recovery infrastructure.

Economic Feasibility

- ✓ Achieving 5x to 50x lower compressor power than state-of-the-art waste heat recovery process



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Toward >100 W/cm^2 2PIC with Lotus-type porous copper and Galden® HT-55

Fanny Oukhemanou PhD

Sustainable Heat Transfer Fluid Program Manager – Syensqo

K. Yuki, T. Okafuji, K. Yuki, - Department of Mechanical Engineering, Tokyo University of Science-Yamaguchi, Sanyo-Onoda

T. Ide, T. Ogushi, M. Murakami - Lotus Thermal Solution Inc., Osaka

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LTS Lotus Thermal Solution

公立大学法人 山陽小野田市立 山口東京理科大学 Sanyo-Onoda City University

SYENSQO

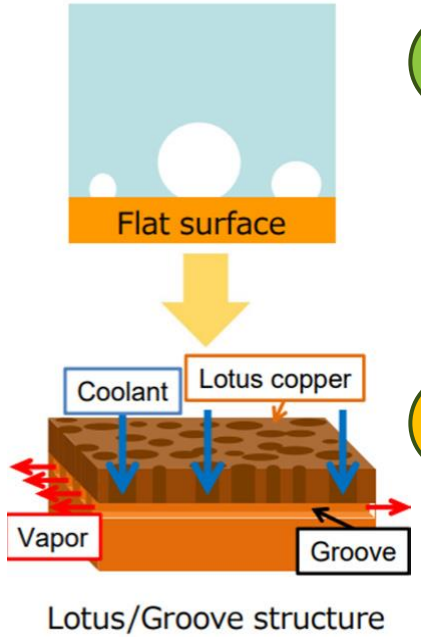
OCP BRONZE Member

How do we bridge the gap to >100 W/cm^2?

- Why >100 W/cm^2 ?
  - Cooling is becoming a limiting factor for dense, serviceable compute platforms as accelerator power keeps rising.
  - Today, air and direct-to-chip cooling dominate; next-gen chips will need immersion cooling (targeted broad adoption in 2028–2030).
  - The current 2PIC setup reaches ~15 W/cm^2: OK for first-gen immersion, but chip heat flux is trending toward ~100 W/cm^2 and beyond.
  - **A ~7× step change is needed:** the boiling surface + fluid package should be designed for ≥100 W/cm^2 to keep margin vs. critical heat flux (CHF).

15 W/cm^2 Planar Cu → 100 W/cm^2 Lotus Cu

- Two independent, combinable levers



Lotus/Groove structure

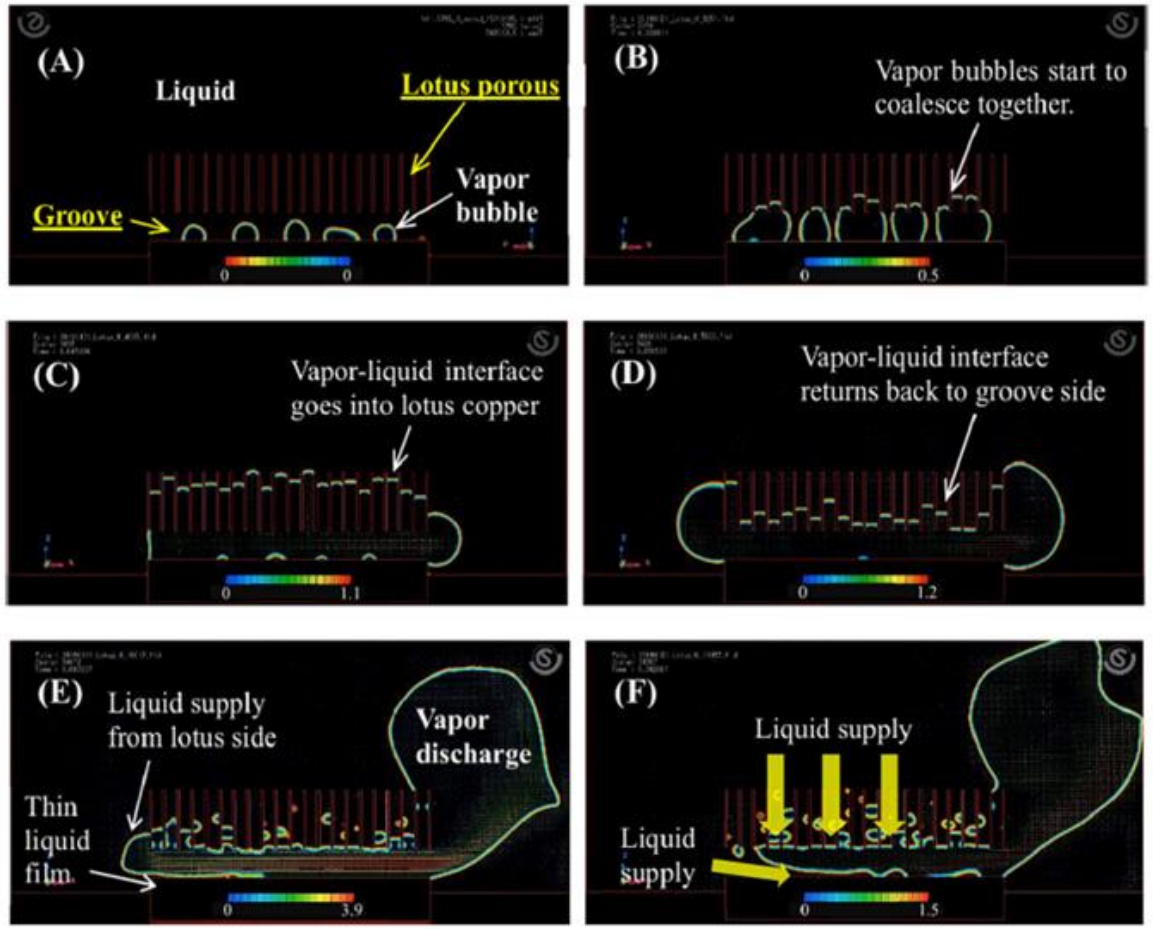
1. Boiling surface morphology — *Planar Cu to Lotus porous Cu*
2. Dielectric fluid — *T_{sat}, Latent heat, Surface tension*

2PIC= 2 phase immersion cooling

The "breathing" mechanism to increase CHF

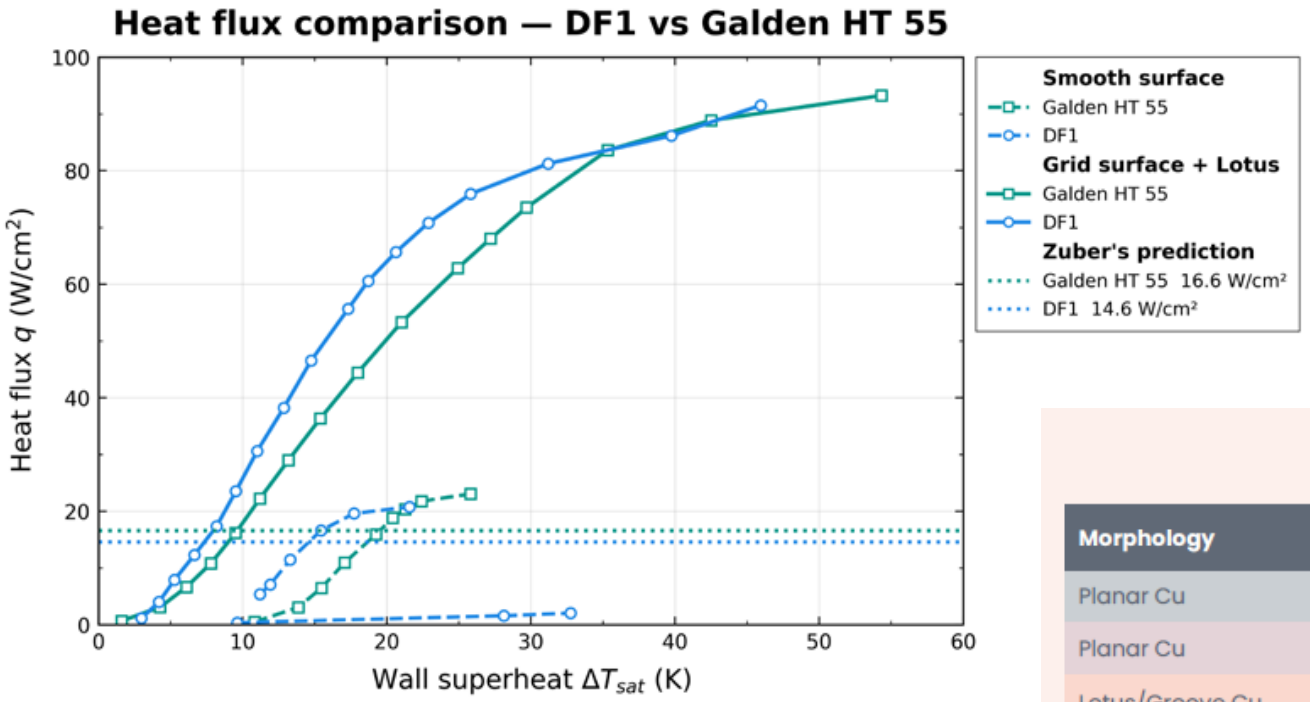
- To increase the Critical Heat Flux (CHF), we need to keep the surface wet

  - Unidirectional millimeter-scale pores provide a dedicated liquid path.
  - Vapor is discharged laterally through the grooves.
  - Flow separation suppresses a coalesced vapor blanket and delays dry-out.
  - Small pores: more nucleation sites at low heat flux.
  - Larger pores: lower vapor resistance at high heat flux.

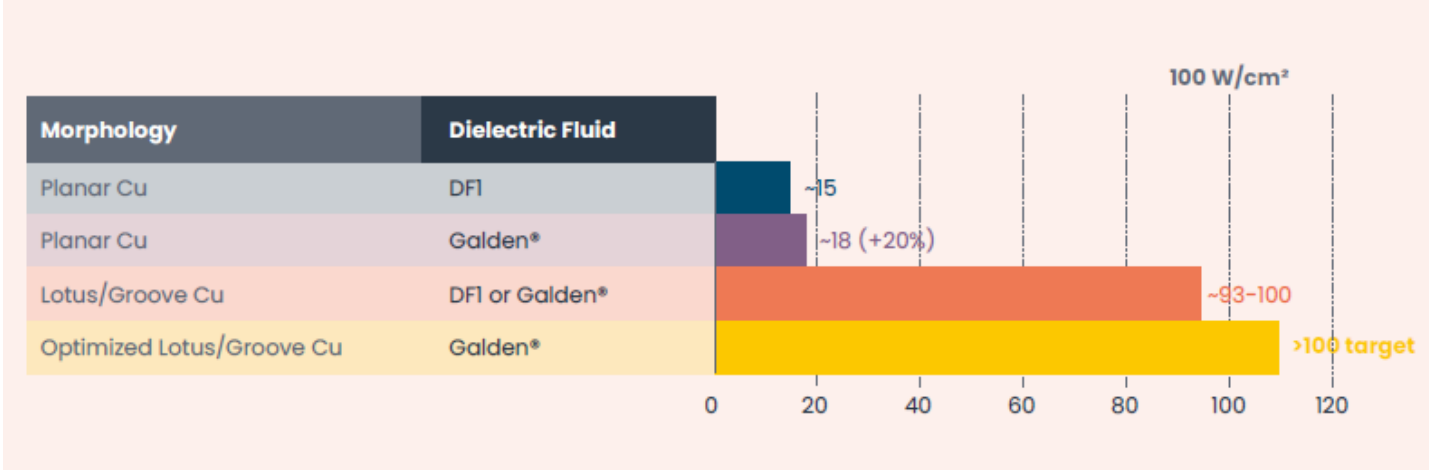


K. Yuki *et al.*, "Critical heat flux enhancement by breathing phenomenon induced by lotus/groove heat transfer surface," Appl. Thermal Eng. 274, 126776 (2025)

Galden® HT-55 + optimized lotus type Cu is a direct route to exceed 100 W/cm²



Fluid adds margin. Surface architecture creates the step change.



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Rack-Scale DI Water 2-Phase Cooling DTC for Next-Gen 1U AI Servers

Ching-chih Wei, Ph.D.

CEO & Co-Founder of CreAlien Co., Ltd.



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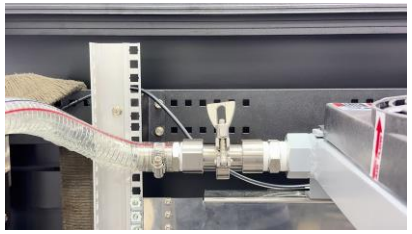
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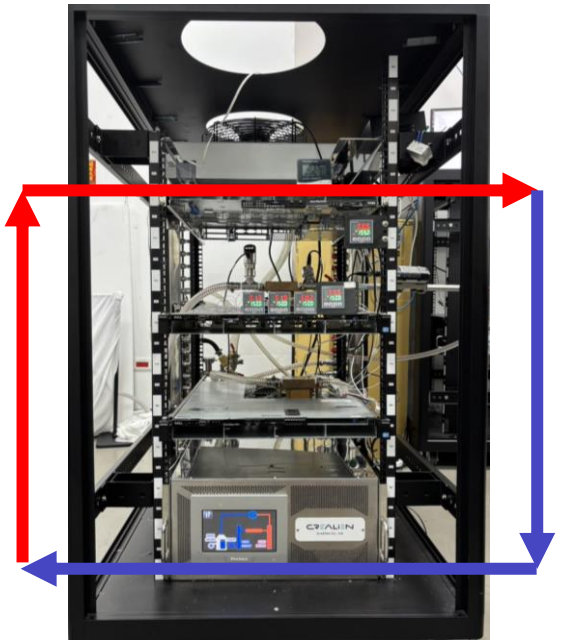
Intro of W2PC

- **Configurable T_{sat} Baseline :** The sub-atmospheric operating pressure dynamically anchors the boiling saturation temperature T_{sat} between 45°C and 60°C, matching varying silicon TDPs and ambient conditions.
- **Tight Secondary Inlet Subcooling :** TCS liquid enters the coldplate at 5–7°C subcooling below T_{sat} , preventing pre-mature flashing.
- **Chiller-less Primary Rejection :** Maintaining a 10 °C approach temperature across the condenser allows direct facility heat rejection via passive dry coolers under ASHRAE W40 to W45 / W+ warm water standards.

ΔT ~10 C

Primary Side Temp.

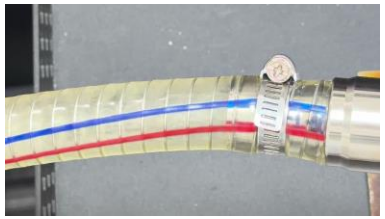




45 C ~ 60 C

Tunable T_{sat}

Vapor cooling



ΔT_{sub} 5 ~ 7 C

Warm Water Inlet

Subcooling

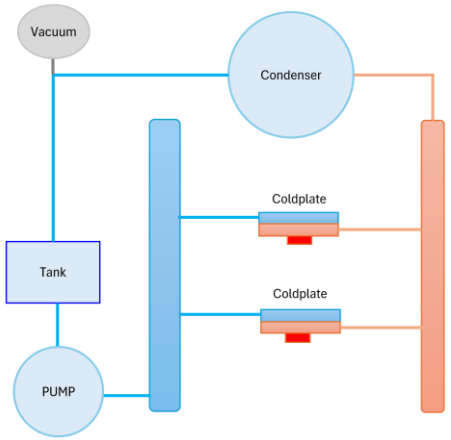
Advantages and Challenges

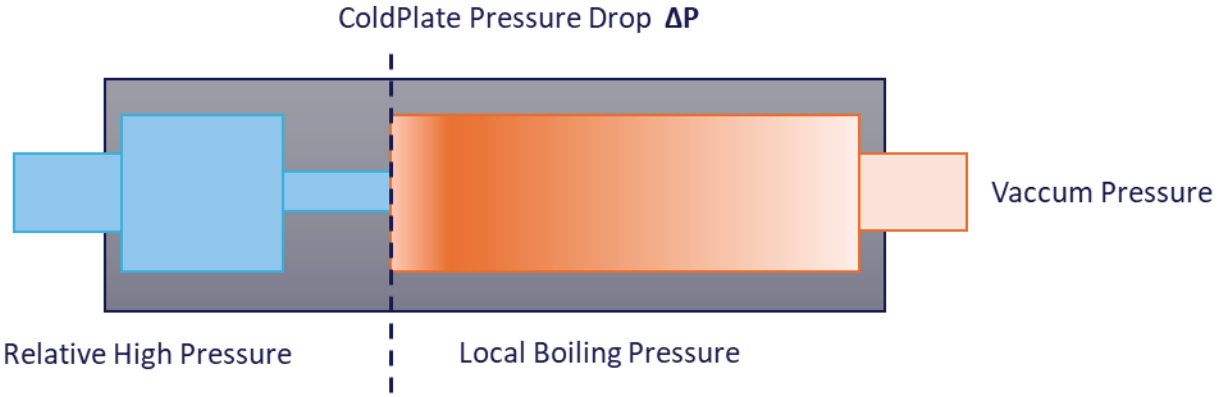
Advantages

- **Drastic Flow Rate Reduction:** Utilizing water's massive enthalpy of vaporization, the required mass flow rate is slashed to just 1/3 of conventional single-phase liquid cooling.
- **Inherently Dry-Out Free:** Operates reliably within a low mass vapor quality regime, preventing liquid film breakdown and eliminating localized dry-out risks.
- **Vacuum-Driven Flow Stability:** The sub-atmospheric pressure differential rapidly evacuates the vapor-liquid mixture downstream, effectively mitigating parallel flow instability and crosstalk across multiple branches.

Challenges

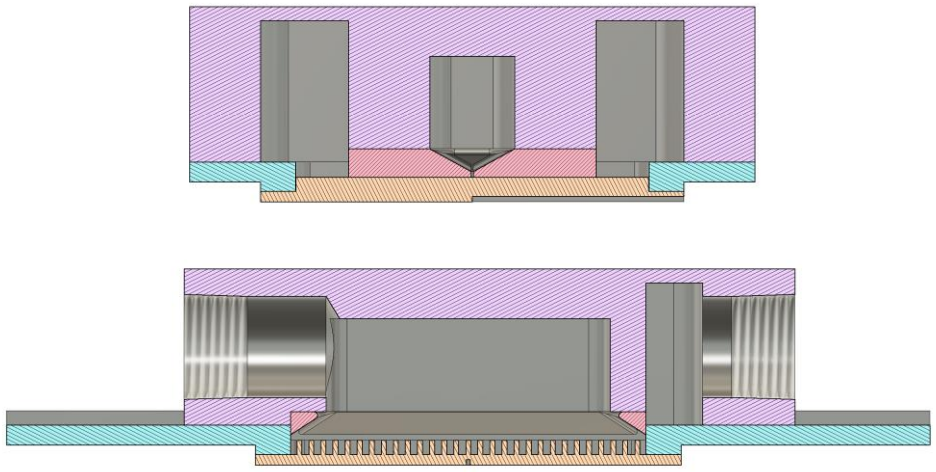
- **Vapor Choking Limit:** Low two-phase sonic velocity under vacuum conditions introduces choking risks; boundary characterization is ongoing to push this operational envelope.
- **High-Density 1U Validation:** Successfully dissipated 2.0 kW over a 50 × 50 mm footprint in a 1U profile, maintaining exceptional baseplate thermal uniformity.
- **Ultra-Slim Scalable Architecture:** Compressed internal coldplate profile down to 13 mm; calibrating the inlet-to-outlet geometric ratio enables direct scaling across diverse accelerator layouts.

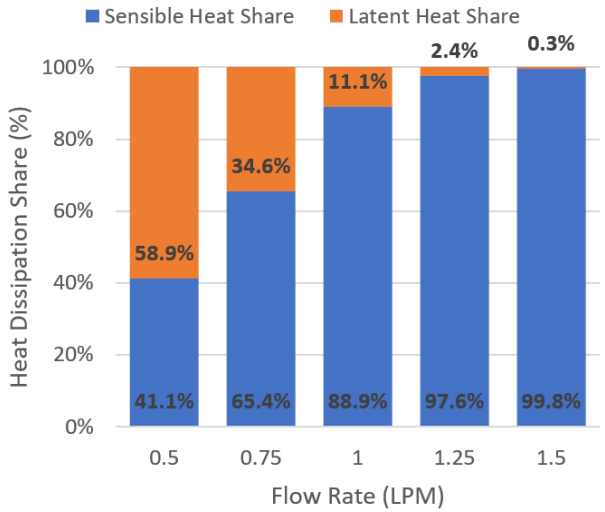


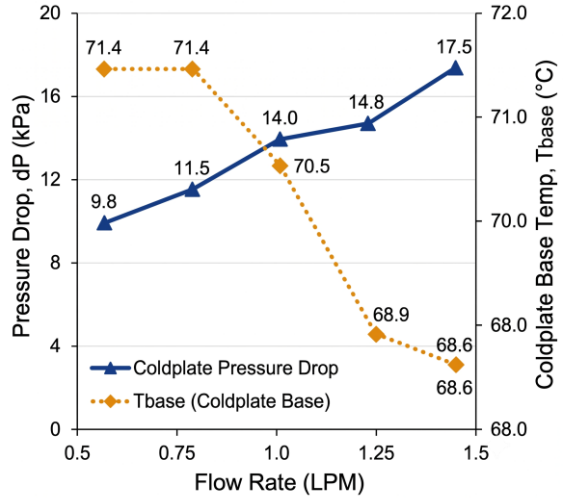


Design and Results of ColdPlate

- Jet-impingement skived-fin architecture within 0.5 mm pitch, engineered for 2.0 kW dissipation over a 50X50 mm footprint.
- Vertical discharge into an upper chamber with a 270X geo-expan reduces peak vapor velocity below the sonic choking limit.
- Calibrated single-phase inlet pressure restrictions isolate node hydraulics from downstream two-phase fluctuations.
- With latent heat accounting for 58.9% of total heat removal, vapor quality remains below 0.06, inherently preventing dry-out.
- Fin pitch down to 0.2 mm achieves T_{case} to 59.5℃ at 2.5 kW, with latent heat accounting for approx. 59% of heat removal at 0.5 LPM.

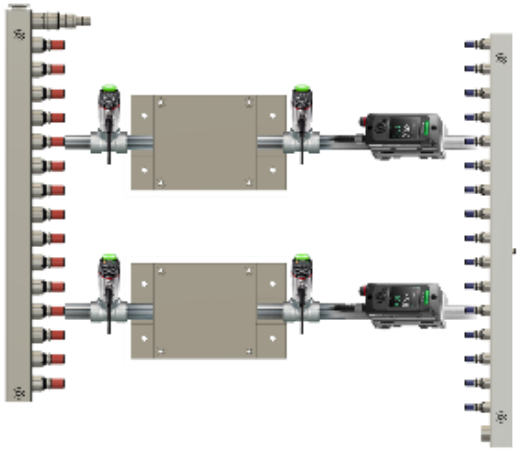




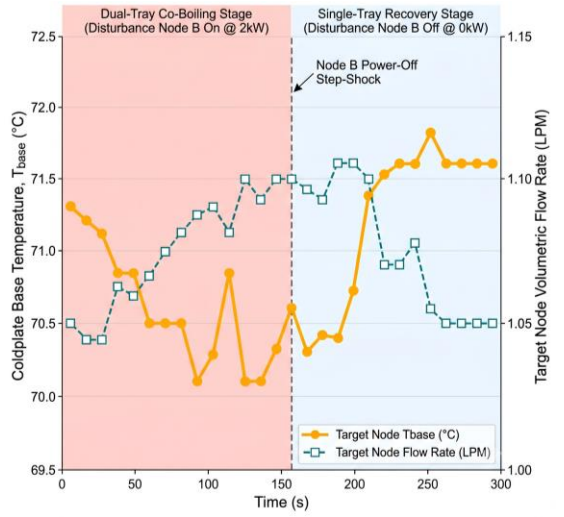


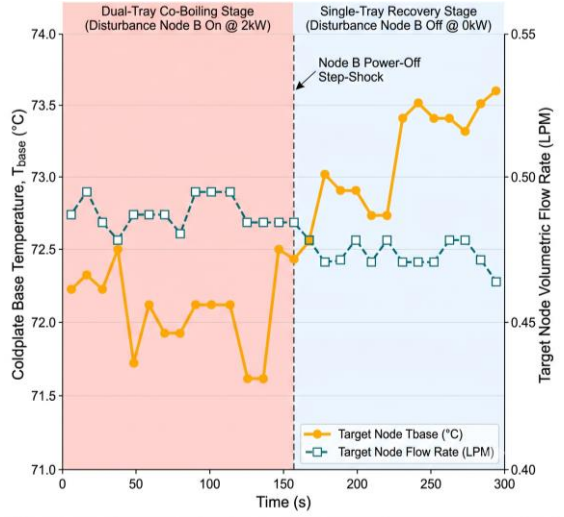
Rack-level Parallel and Stabilization Test

- **Dynamic Load-Shedding Test:** Simulated 0kW to 2.0 kW step-shocks across parallel nodes at 1.0 LPM and 0.5 LPM.
- **No Fluid Robbing:** Transient flow variance restricted to approx. 6%, effectively isolating multi-tray hydraulic crosstalk.
- **Clamped Thermal Response:** Suppressed temperature fluctuation to $\Delta T \leq 2$ ℃, eliminating thermal runaway.
- **Thermodynamic Damping:** Sub-atmospheric saturation matrix inherently stabilizes the cluster against extreme workload fluctuations.



Item	Spec & Data	
Node A Power	2000W > 0W	
Node B Power	2000W	
Flow Rate	0.5 LPM	1.0 LPM
Flow Rate Diff.	0.03 LPM	0.09 LPM
Tcase Diff.	~ 2C	< 2C





Next Action

- **In-Rack Field Validation:**
  On-site PoS field trial currently underway, targeting Q4 milestone completion.

- **In-Row Architecture Scaling:**
  Completing internal validation and stress testing for the 200 kW+ in-row system in Q4.

- **Call for In-Row PoC:**
  Actively seeking data center and industry partners for on-site In-Row PoC deployments and joint certification.

- **Vacuum Safety & Standards:**
  Establishing standardized vacuum leak-containment thresholds and fail-safe operational criteria in Q4.

- **Long-Term Fluid Reliability:**
  Ongoing microbiological growth analysis and DI water chemistry monitoring under continuous operation, targeting Q4 milestone completion.

OCP FUTURE TECHNOLOGIES SYMPOSIUM

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